

# Ángel López-Oriona

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🌐 [www.lopezoriona.com](http://www.lopezoriona.com)



## Work Experience

- Sep 2026 – **Assistant Professor**, ZAYED UNIVERSITY, Dubai, United Arab Emirates.  
Present
- Sep 2023 – **Postdoctoral Research Fellow**, KING ABDULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY (KAUST), Thuwal, Saudi Arabia.  
Aug 2026
- May 2020 – **PhD Candidate**, UNIVERSITY OF A CORUÑA, A Coruña, Spain.  
Jul 2023
- Jan 2021 – **Instructor**, EUROPEAN CENTER FOR POSTGRADUATE STUDIES (CEMP), A Coruña, Spain.  
Dec 2021
- Jun 2019 – **Data Scientist**, ESTRELLA GALICIA, A Coruña, Spain.  
Feb 2020
- Before 2019 **Online Poker Player**, SELF-EMPLOYED.

## Education

- PhD in Statistics.** UNIVERSITY OF A CORUÑA, A Coruña, Spain. Cum Laude Mention, International Mention, and Extraordinary Doctoral Award.
- Master of Science in Big Data Analytics.** EUROPEAN UNIVERSITY OF MADRID, Madrid, Spain. With Honors.
- Master of Science in Statistical Techniques.** UNIVERSITY OF SANTIAGO DE COMPOSTELA, Santiago de Compostela, Spain. With Honors.
- Bachelor of Science in Mathematics.** UNIVERSITY OF SANTIAGO DE COMPOSTELA, Santiago de Compostela, Spain.

## Journal Papers

- 2026 **López-Oriona, Á.**, Sun, Y., & Vilar, J. A. (2026). Clustering locally stationary time series using quantile autocorrelations. *Journal of Computational and Graphical Statistics*, 35(3), 1376–1396.
- 2026 Ma, Z., **López-Oriona, Á.**, Sun, Y., & Ombao, H. (2026). ROBCPCA: A robust multivariate time series clustering method based on common principal component analysis. *Journal of Classification*. <https://doi.org/10.1007/s00357-026-09536-7>.
- 2026 **López-Oriona, Á.**, & Sun, Y. (2026). Forecasting time series collections via fuzzy clustering. *Fuzzy Sets and Systems*, 530, 109748.
- 2025 Ma, Z., **López-Oriona, Á.**, Sun, Y., & Ombao, H. (2025). FCPCA: Fuzzy clustering of high-dimensional time series based on common principal component analysis. *International Journal of Approximate Reasoning*, 187, 109552.

- 2025 **López-Oriona, Á.**, & Sun, Y. (2025). Lag selection in feature-based clustering of time series. *Knowledge-Based Systems*, 329(A), 114258.
- 2025 **López-Oriona, Á.**, Sun, Y., & Vilar, J. A. (2025). Improving the prediction accuracy of statistical models: A new hierarchical clustering approach. *Statistics and Computing*, 35, 168.
- 2025 **López-Oriona, Á.**, Montero-Manso, P., & Vilar, J. A. (2025). Time series clustering based on prediction accuracy of global forecasting models. *Knowledge-Based Systems*, 323, 113649.
- 2025 **López-Oriona, Á.**, Sun, Y., & Shang, H. L. (2025). Dependence-based fuzzy clustering of functional time series. *Journal of Computational and Graphical Statistics*, 34(4), 1729–1741.
- 2025 **López-Oriona, Á.**, Sun, Y., & Crujeiras, R. M. (2025). Fuzzy clustering of circular time series with applications to wind data. *Environmetrics*, 36(2), e2902.
- 2024 **López-Oriona, Á.**, & Vilar, J. A. (2024). Analyzing categorical time series with the R package *ctsfeatures*. *Journal of Computational Science*, 76, 102233.
- 2023 **López-Oriona, Á.**, Weiss, C. H., & Vilar, J. A. (2023). Two novel distances for ordinal time series and their application to fuzzy clustering. *Fuzzy Sets and Systems*, 468, 108590.
- 2023 Trigo-Tasende, N., et al. (2023). Wastewater early warning system for SARS-CoV-2 outbreaks and variants in a Coruña, Spain. *Environmental Science and Pollution Research*, 30, 79315-79334.
- 2023 **López-Oriona, Á.**, & Vilar, J. A. (2023). Ordinal Time Series Analysis with the R Package *otsfeatures*. *Mathematics*, 11(11), 2565.
- 2023 **López-Oriona, Á.**, & Vilar, J. A. (2023). Machine learning for multivariate time series with the R package *mlmts*. *Neurocomputing*, 150, 55-82.
- 2023 **López-Oriona, Á.**, Vilar, J. A., & D'Urso, P. (2023). Hard and soft clustering of categorical time series based on two novel distances with an application to biological sequences. *Information Sciences*, 624, 467-492.
- 2022 **López-Oriona, Á.**, & Vilar, J. A. (2022). The bootstrap for testing the equality of two multivariate time series with an application to financial markets. *Information Sciences*, 616, 255-275.
- 2022 **López-Oriona, Á.**, D'Urso, P., Vilar J. A., & Lafuente-Rego, B. (2022). Quantile-based fuzzy C-means clustering of multivariate time series: Robust techniques. *International Journal of Approximate Reasoning*, 150, 55-82.
- 2022 **López-Oriona, Á.**, Vilar, J. A., & D'Urso, P. (2022). Quantile-based fuzzy clustering of multivariate time series in the frequency domain. *Fuzzy Sets and Systems*, 443(B), 115-154.
- 2022 **López-Oriona, Á.**, D'Urso, P., Vilar J. A., & Lafuente-Rego, B. (2022). Spatial weighted robust clustering of multivariate time series based on quantile dependence with an application to mobility during COVID-19 pandemic. *IEEE Transactions on Fuzzy Systems*, 30(9), 3990-4004.
- 2022 Vallejo, Juan A., et al. (2022). Modeling the number of people infected with SARS-COV-2 from wastewater viral load in Northwest Spain. *Science of The Total Environment*, 811, 152334.
- 2021 **López-Oriona, Á.**, & Vilar, J. A. (2021). F4: An all-purpose tool for multivariate time series classification. *Mathematics*, 9(23), 3051.
- 2021 **López-Oriona, Á.**, & Vilar, J. A. (2021). Outlier detection for multivariate time series: A functional data approach. *Knowledge-Based Systems*, 233, 107527.
- 2021 **López-Oriona, Á.**, & Vilar, J. A. (2021). Quantile cross-spectral density: A novel and effective tool for clustering multivariate time series. *Expert Systems with Applications*, 185, 115677.
- 2020 **López-Oriona, Á.**, Ling, J. F., & Sellero, C. S. (2020). Photocenter shift effect in double stars of Gaia DR2 database. *Acta Astronomica*, 70(1), 19-32.

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## Submitted Papers and Works in Progress

- 2026 **López-Oriona, Á.**, Sun, Y., & Shang, H. L. (2026). White noise testing for functional time series via functional quantile autocorrelation. <https://arxiv.org/pdf/2601.09371>. Under review in *Journal of Computational and Graphical Statistics*.
- 2026 **López-Oriona, Á.**, & Sun, Y. (2026). Amortized neural clustering of time series based on statistical features. <https://arxiv.org/pdf/2605.13128>. Under review in *Statistics and Computing*.
- 2026 **López-Oriona, Á.**, & Vilar, J. A. (2026). Nonlinear tests of independence for circular time series. Under review in *Journal of Time Series Analysis*.
- 2026 **López-Oriona, Á.**, & Matabuena, Marcos. (2026). Digital health biomarkers based on probability distributions of time series via bayesian context tree representations. In preparation.
- 2026 **López-Oriona, Á.**, & Ambriz-Lobato, I. E. (2026). Partial quantile spectral measures with financial applications. In preparation.
- 2026 **López-Oriona, Á.**, & Jiménez-Varón, C. F. (2026). An amortized neural framework for forecasting and inference in functional time series. In preparation.
- 2026 **López-Oriona, Á.** (2026). Towards amortized fuzzy clustering of time series. In preparation.
- 2026 Ma, Z., **López-Oriona, Á.**, Sun, Y., & Ombao, H. (2026). Anomaly-aware robust fuzzy subspace clustering for high-dimensional multivariate time series. <https://arxiv.org/pdf/2510.26982>. Under review in *Signal Processing*. Second review round.
- 2026 Ma, Z., **López-Oriona, Á.**, Sun, Y., & Ombao, H. (2026). Forecasting multivariate time series under predictive heterogeneity: a validation-driven clustering framework. <https://arxiv.org/pdf/2604.13748>. Under review in *International Journal of Forecasting*.
- 2026 Ma, Z., **López-Oriona, Á.**, Jiang, J., Sun, Y., & Ombao, H. (2026). Adaptive multi-scale forecasting and gate-localized conformal prediction for multivariate nonstationary time series. Under review in *AAAI Conference on Artificial Intelligence*. Second review round.

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## Conference Papers

- 2023 **López-Oriona, Á.**, Montero-Manso, P., & Vilar, J. A. (2023). Clustering of time series based on forecasting performance of global models. *Advanced Analytics and Learning on Temporal Data: 7th ECML PKDD Workshop, AALTD 2022, Grenoble, France, September 19–23, 2022, Revised Selected Papers*, (pp. 18-33). Springer International Publishing.
- 2023 **López-Oriona, Á.**, Vilar, J. A., & D'Urso, P. (2023). Unsupervised classification of categorical time series through innovative distances. *17th Conference of the International Federation of Classification Societies, IFCS 2022, Porto, Portugal, July 19–23, 2022*, (pp. 233-241). Springer International Publishing.
- 2022 **López-Oriona, Á.**, & Vilar, J. A. (2022). The bootstrap for testing the equality of two multivariate stochastic processes with an application to financial markets. *Engineering Proceedings*, 18(1), 38.
- 2021 **López-Oriona, Á.**, D'Urso, P., Vilar, J. A., & Lafuente-Rego, B. (2021). Robust methods for soft clustering of multidimensional time series. *Engineering Proceedings*, 7(1), 60.

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## Teaching

### Zayed University

- Fall 2026 MTH-281-205. *Probability and Statistics*. Undergraduate.
- Fall 2026 MTH-281-204. *Probability and Statistics*. Undergraduate.
- Fall 2026 MTH-213-207. *Business Statistics*. Undergraduate.
- Fall 2026 MTH-213-203. *Business Statistics*. Undergraduate.

### European Center for Postgraduate Studies (CEMP)

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## Advising and Mentoring

### PhD advisees

- Aug 2023 - Ziling Ma, King Abdullah University of Science and Technology (KAUST). *Robust and fuzzy methods for high-dimensional time series clustering and forecasting*. Co-advised with Professor Present Ying Sun (KAUST) and Prof. Hernando Ombao (KAUST).

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## Conference Proceedings

- 2023 **López-Oriona, Á.**, Weiss, C. H., & Vilar, J. A. (2023), *Fuzzy clustering of ordinal time series based on two novel distances*. 5th International Conference on Statistics: Theory and Applications (ICSTA 2023), London, United Kingdom, 03-05 August 2023. [https://avestia.com/ICSTA2023\\_Proceedings/files/paper/ICSTA\\_110.pdf](https://avestia.com/ICSTA2023_Proceedings/files/paper/ICSTA_110.pdf).
- 2022 **López-Oriona, Á.**, & Vilar, J. A. (2022), *Machine learning for multivariate time series with the R package mlmts*. Joint Statistical Meetings (JSM 2022), Washington, D.C., USA, 06-11 August 2022. [www2.amstat.org/meetings/Proceedings/index.cfm](http://www2.amstat.org/meetings/Proceedings/index.cfm).
- 2022 **López-Oriona, Á.**, Vilar, J. A., & D'Urso, P. (2022), *Unsupervised classification of categorical time series through innovative distances*. 4th International Conference on Statistics: Theory and Applications (ICSTA 2022), Prague, Czech Republic, 28-30 July 2022. [https://avestia.com/ICSTA2022\\_Proceedings/files/paper/ICSTA\\_111.pdf](https://avestia.com/ICSTA2022_Proceedings/files/paper/ICSTA_111.pdf).
- 2021 **López-Oriona, Á.**, D'Urso, P., Vilar, J. A., & Lafuente-Rego, B. (2021), *Spatial weighted robust clustering of multivariate time series based on quantile dependence with an application to mobility during COVID-19 pandemic*. 13th International Workshop on Fuzzy Logic and Applications (WILF 2021), Vietri sul Mare, Italy, 20-22 December 2021. <http://ceur-ws.org/Vol-3074/paper07.pdf>.
- 2021 **López-Oriona, Á.**, & Vilar, J. A. (2021), *An effective tool for clustering multivariate time series with an application to financial markets*. XV Galician Conference on Statistics and Operations Research (SGAPEIO 2021), Santiago de Compostela, Spain, 04-06 November 2021. <http://xvcongreso.sgapeio.es/descargas/actas-xvSGAPEIO.pdf>.

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## Conference Oral Presentations

- 2026 New Frontiers in Statistical Inference on Geometry, Topology and Time Series, Kyoto, Japan: *Time series clustering: pattern recognition, forecasting, and amortized inference*. Invited.
- 2025 8th International Conference on Econometrics and Statistics (EcoSta 2025), Tokyo, Japan: *Clustering locally stationary time series using quantile autocorrelations*. Invited.
- 2024 26th International Conference on Computational Statistics (COMPSTAT 2024), Giessen, Germany: *Fuzzy clustering of circular time series based on a new dependence measure with applications to wind data*. Invited.
- 2024 7th International Conference on Econometrics and Statistics (EcoSta 2024), Beijing, China: *Fuzzy clustering of circular time series based on a novel distance with an application to wind data*. Invited.
- 2023 Australian Statistical Conference (ASC 2023), Wollongong, Australia: *Fuzzy clustering of circular time series based on a novel distance with an application to wind data*. Contributed.
- 2023 Joint Statistical Meetings (JSM 2023), Toronto, Canada: *Fuzzy clustering of ordinal time series based on two novel distances with financial applications*. Contributed.
- 2023 5th International Conference on Statistics: Theory and Applications (ICSTA 2023), London, United Kingdom: *Fuzzy clustering of ordinal time series based on two novel distances*. Contributed.
- 2023 9th International Conference on Time Series and Forecasting (ITISE 2023), Gran Canaria, Spain: *Clustering of time series based on forecasting performance of global models*. Contributed.

- 2023 9th International Conference on Time Series and Forecasting (ITISE 2023), Gran Canaria, Spain: *Machine learning for multivariate time series with the R package **mlmts***. Contributed.
- 2022 15th International Conference of the ERCIM WG on Computational and Methodological Statistics (CMStatistics 2022), London, United Kingdom: *Time series clustering based on prediction accuracy of global forecasting models*. Invited.
- 2022 International Conference on Mathematics, Computational Sciences and Statistics (ICoMCoS 2022), Surabaya, Indonesia: *The bootstrap for testing the equality of two multivariate stochastic processes with an application to financial markets*. Contributed.
- 2022 International Conference on Mathematics, Computational Sciences and Statistics (ICoMCoS 2022), Surabaya, Indonesia: *Unsupervised learning of temporal data based on global prediction models*. Contributed.
- 2022 7th Workshop on Advanced Analytics and Learning on Temporal Data (AALTD 2022), Grenoble, France: *Time series clustering based on prediction accuracy of global forecasting models*. Contributed.
- 2022 Joint Statistical Meetings (JSM 2022), Washington, D.C., USA: *Machine learning for multivariate time series with the R package **mlmts***. Contributed.
- 2022 4th International Conference on Statistics: Theory and Applications (ICSTA 2022), Prague, Czech Republic: *Unsupervised classification of categorical time series through innovative distances*. Contributed.
- 2022 17th Conference of the International Federation of Classification Societies (IFCS 2022), Porto, Portugal: *Unsupervised classification of categorical time series through innovative distances*. Contributed.
- 2022 8th International Conference on Time Series and Forecasting (ITISE 2022), Gran Canaria, Spain: *The bootstrap for testing the equality of two multivariate stochastic processes with an application to financial markets*. Contributed.
- 2022 XXIII International Symposium of Mathematical Methods Applied to Sciences (SIMMAC 2022), San José, Costa Rica: *Quantile-based fuzzy clustering of multivariate time series in the frequency domain*. Contributed.
- 2022 11th Conference of the Asian Regional Section of the International Association for Statistical Computing (IASC-ARS 2022), Kyoto, Japan: *Soft clustering of multidimensional time series*. Contributed.
- 2021 13th International Workshop on Fuzzy Logic and Applications (WILF 2021), Vietri sul Mare, Italy: *Spatial weighted robust clustering of multidimensional time series based on quantile dependence with an application to mobility during COVID-19 pandemic*. Contributed.
- 2021 14th International Conference of the ERCIM WG on Computational and Methodological Statistics (CMStatistics 2021), London, United Kingdom: *Spatial weighted robust clustering of multivariate time series with an application to COVID-19 pandemic*. Invited.
- 2021 XV Galician Conference on Statistics and Operations Research (SGAPEIO 2021), Santiago de Compostela, Spain: *An effective tool for clustering multivariate time series with an application to financial markets*. Contributed.
- 2021 13th International Conference on Fuzzy Computation Theory and Applications (FCTA 2021), Virtual: *Fuzzy clustering of multivariate time series based on quantile dependence*. Contributed.
- 2021 XXV Congress of the Portuguese Statistical Society (SPE 2021), Évora, Portugal: *Quantile-based fuzzy C-means clustering of multivariate time series: Robust techniques*. Contributed.
- 2021 4th XoveTIC Conference for young researchers (XoveTIC 2021), A Coruña, Spain: *Quantile-based fuzzy C-means clustering of multivariate time series: Robust techniques*. Contributed.
- 2021 7th International Conference on Time Series and Forecasting (ITISE 2021), Gran Canaria, Spain: *F4: An all-purpose tool for multivariate time series classification*. Contributed.

- 2020 13th International Conference of the ERCIM WG on Computational and Methodological Statistics (CMStatistics 2020), London, United Kingdom: *A novel structure-based approach for multivariate time series clustering*. Invited.

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## Conference Poster Presentations

- 2026 XXV International Conference on Robust Statistics (ICORS 2026), Istanbul, Turkey: *Amortized neural clustering of time series based on statistical features*. Contributed.
- 2026 XVII Latin American Congress of Probability and Mathematical Statistics (CLAPEM 2026), Montevideo, Uruguay: *Amortized neural clustering of time series based on statistical features*. Contributed.
- 2025 The KAUST 2025 Workshop on Statistics, Thuwal, Saudi Arabia: *Amortized neural clustering of time series based on statistical features*. Contributed.
- 2024 The KAUST 2024 Workshop on Statistics, Thuwal, Saudi Arabia: *Dependence-based fuzzy clustering of functional time series*. Contributed.
- 2023 The KAUST 2023 Workshop on Statistics, Thuwal, Saudi Arabia: *Time series clustering based on prediction accuracy of global forecasting models*. Contributed.
- 2021 5th International Workshop on Functional and Operatorial Statistics (IWFOS 2021), Brno, Czech Republic: *Outlier detection for multivariate time series: a functional data approach*. Contributed.

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## Conference Contributions as Coauthor

- 2026 9th International Conference on Econometrics and Statistics (EcoSta 2026), Kyoto, Japan: *Adaptive multi-scale forecasting and gate-localized conformal prediction for multivariate nonstationary time series*. Speaker: Ziling Ma.
- 2026 New Frontiers in Statistical Inference on Geometry, Topology and Time Series, Kyoto, Japan: *Forecasting multivariate time series under predictive heterogeneity: A validation-driven clustering framework*. Speaker: Ziling Ma.
- 2026 XXV International Conference on Robust Statistics (ICORS 2026), Istanbul, Turkey: *Forecasting multivariate time series under predictive heterogeneity: A validation-driven clustering framework*. Speaker: Ziling Ma.
- 2025 IMS International Conference on Statistics and Data Science (ICSIDS 2025), Sevilla, Spain: *Robust fuzzy clustering for high-dimensional multivariate time series with outlier detection*. Speaker: Ziling Ma.
- 2025 19th International Joint Conference on Computational and Financial Econometrics and Computational and Methodological Statistics (CFE-CMStatistics 2025), London, United Kingdom: *Robust fuzzy clustering for high-dimensional multivariate time series with outlier detection*. Speaker: Ziling Ma.
- 2025 8th International Conference on Econometrics and Statistics (EcoSta 2025), Tokyo, Japan: *FPCA: Fuzzy clustering of high-dimensional MTS based on common principal component analysis with robust extensions*. Speaker: Ziling Ma.
- 2025 24th International Conference on Robust Statistics (ICORS 2025), Stresa, Italy: *Fuzzy clustering of multivariate time series based on common principal component analysis with robust extensions*. Speaker: Ziling Ma.
- 2022 17th Conference of the International Federation of Classification Societies (IFCS 2022), Porto, Portugal: *Clusters based on prediction accuracy of global forecasting models*. Speaker: Pablo Montero-Manso.
- 2022 42nd International Symposium on Forecasting (ISF 2022), Oxford, United Kingdom: *Improving the forecasting accuracy of global models/cross-learning in large datasets by finding clusters of similar time series*. Speaker: Pablo Montero-Manso.



- 2022 32nd European Congress of Clinical Microbiology and Infectious Diseases (ECCMID 2022), Lisbon, Portugal: *COVIDBENS: wastewater-based epidemiology to monitor COVID-19 pandemic and to predict new outbreaks in A Coruña (NW Spain)*. Speaker: Margarita Poza Domínguez.

## Conference Organizations

- 2025 8th International Conference on Econometrics and Statistics (EcoSta 2025), Tokyo, Japan. Organizer of Session EO147, entitled *Statistical learning for time series*.
- 2024 7th International Conference on Econometrics and Statistics (EcoSta 2024), Beijing, China. Organizer of Session EO317, entitled *Clustering and classification for time series*.
- 2021 13th International Conference on Fuzzy Computation Theory and Applications (FCTA 2021), Virtual. Organizer of Session 1A (poster presentations).

## Seminars and Short Courses

- 2026 Academic seminar for the Computer, Electrical and Mathematical Sciences and Engineering (CEMSE) Division, King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia: *Time series clustering: pattern recognition, forecasting, and amortized inference*. Invited by Prof. Ying Sun.
- 2024 Practical tutorial for the Biostatistics Group, King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia: *Building R packages. From the first steps to CRAN submission*. Invited by Prof. Hernando Ombao.
- 2024 Academic seminar for the Sequence Analysis Association (SAA), Stony Brook University, New York, USA: *Analyzing categorical time series with the R package ctsfeatures*. Invited by Prof. Tim Liao.
- 2024 Academic seminar for the Computer, Electrical and Mathematical Sciences and Engineering (CEMSE) Division, King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia: *Fuzzy clustering of circular time series based on a new distance with applications to wind data*. Invited by Prof. Paula Moraga.
- 2022 Academic seminar for the Department of Mathematics and Statistics, Helmut Schmidt University, Hamburg, Germany: *Clustering of categorical time series through innovative distances*. Invited by Prof. Christian H. Weiss.
- 2022 Academic seminar for the Discipline of Business Analytics, the University of Sydney, Business School, Sydney, Australia: *Time series clustering based on prediction accuracy of global forecasting models*. Invited by Prof. Pablo Montero-Manso.

## Software

**R package mlmts:** López-Oriona, Á., & Vilar, J. A. mlmts: Machine Learning Algorithms for Multivariate Time Series, r package version 1.1.2 (2024). URL <https://CRAN.R-project.org/package=mlmts>.

**R package ctsfeatures:** López-Oriona, Á., & Vilar, J. A. ctsfeatures: Analyzing Categorical Time Series, r package version 1.2.2 (2024). URL <https://CRAN.R-project.org/package=ctsfeatures>.

**R package otsfeatures:** López-Oriona, Á., & Vilar, J. A. otsfeatures: Ordinal Time Series Analysis, r package version 1.0.0 (2023). URL <https://CRAN.R-project.org/package=otsfeatures>.

## Research Stays

PhD stays

- 2023 Lancaster University. Department of Mathematics and Statistics. Lancaster, United Kingdom.
  - One-month research stay working on clustering of nonstationary time series under the supervision of lecturer Carolina Euan (from 10/06/2023 to 10/07/2023).
- 2022 Helmut Schmidt University. Department of Mathematics and Statistics. Hamburg, Germany.
  - Three-month research stay working on data mining for ordinal time series under the supervision of Prof. Christian H. Weiss (from 20/09/2022 to 20/12/2022).
- 2022 The University of Sydney, Business School. Discipline of Business Analytics. Sydney, Australia.
  - Three-month research stay working on time series clustering and forecasting under the supervision of Prof. Pablo Montero-Manso (from 01/04/2022 to 30/06/2022).
- 2021 Sapienza University of Rome, Faculty of Statistics. Department of Social and Economic Sciences. Rome, Italy.
  - Three-month research stay working on fuzzy clustering of multivariate time series under the supervision of Prof. Pierpaolo D'Urso (from 05/05/2021 to 05/08/2021).

## Courses and Workshops

*A Tutorial on Conformal Prediction.* Instructors: Dr. Anastasios N. Angelopoulos and Dr. Stephen Bates, University of California, Berkeley, United States.

*Statistical Theory.* Instructor: Prof. Noah Simon, University of Washington, Seattle, United States.

*Stochastic Approximation beyond Gradient for Machine Learning: Taming Noise with Guarantees.* Instructor: Prof. Gersende Fort, Laboratoire d'Analyse et d'Architecture des Systèmes, Toulouse, France.

*Short Course on Copulas.* Instructor: Dr. Kiran Karra, Johns Hopkins University, Baltimore, United States.

*Multivariable Calculus.* Instructor: Dr. Julian Grossmann, Self-employed mathematician (formerly Hamburg University of Technology, Hamburg, Germany).

*Measure Theory.* Instructor: Dr. Julian Grossmann, Self-employed mathematician (formerly Hamburg University of Technology, Hamburg, Germany).

*Fundamentals of Probability.* Instructors: Prof. John Tsitsiklis and Prof. Patrick Jaillet, Massachusetts Institute of Technology (MIT), Cambridge, United States.

*Neural Methods for Amortised Inference.* Instructor: Prof. Andrew Zammit Mangion, University of Wollongong, Wollongong, Australia.

*Data Analysis on the Sphere.* Instructor: Prof. Zubair Khalid, Lahore University of Management Sciences, Lahore, Pakistan.

*An Introduction to Hidden Markov Models.* Instructor: Prof. Francesco Lagona, University Roma Tre, Rome, Italy.

## Awards and Honors

- 2024 Extraordinary Doctoral Award by the University of A Coruña (UDC) for the most outstanding PhD thesis defended during the academic year 2022-2023 in the fields of Computer Science and Mathematics.
- 2022 Award by the International Federation of Classification Societies (IFCS) for the best work presented by a PhD student at the 17th Conference of the International Federation of Classification Societies (IFCS 2022).
- 2021 Young researcher grant awarded by the European Society for Fuzzy Logic and Technology (EUSFLAT) for the best work presented by a PhD student at the 13th International Workshop on Fuzzy Logic and Applications (WILF 2021).



- 2020 Award by the company *International Business Machines* (IBM) for the development of the most accurate Deep Learning model in the 1st IBM Competition of Computer Vision held in the European University of Madrid.
- 2020 Award (shared with many colleagues) by the Royal Galician Academy of Sciences (RAGC) for the development of a project (called COVIDBENS) for analyzing and monitoring the COVID-19 pandemic in the area of A Coruña.
- 2019 Merit award for academic achievements in Master's Degree at the University of Santiago de Compostela.

## Fellowships and Mobility Grants

- 2023 Competitive grant for a research stay at Lancaster University awarded by the Research Center for Information and Communication Technologies (CITIC) of the University of A Coruña.
- 2022 Competitive grant for a research stay at the Helmut Schmidt University of Hamburg awarded by the company Inditex in collaboration with the University of A Coruña.
- 2022 Competitive grant for a research stay at the University of Sydney awarded by the Research Center for Information and Communication Technologies (CITIC) of the University of A Coruña.
- 2021 Competitive grant for a research stay at Sapienza University of Rome awarded by the Xunta de Galicia.
- 2020 Competitive fellowship for PhD students (2020-2023) awarded by the Xunta de Galicia.
- 2020 Competitive fellowship for PhD students (2020-2022) awarded by the Center for Information and Communications Technology Research (CITIC) of the University of A Coruña.

## Research Grants

- 01/09/2024 **Statistical inference using flexible methods for complex data: theory and applications.**  
-  
31/08/2027
  - Granting institution: Spanish Ministry of Science and Innovation.
  - Duration: 3 years.
  - Amount: €237,500.
  - Principal investigators: Mario Francisco Fernández and Ricardo Cao Abad.
  - Code: PID2023-147127OB-I00.
  - Role: External Research Collaborator.
- 20/09/2023 **Research methods for environmental statistics.**  
-  
19/09/2026
  - Granting institution: King Abdullah University of Science and Technology (KAUST).
  - Duration: 3 years.
  - Amount: Private.
  - Principal investigator: Ying Sun.
  - Code: BAS/1/1655-01-01.
  - Role: Postdoctoral Research Fellow.

- 01/09/2021 **Flexible statistical methods in data science for complex and big data: theory and applications.**
- 
- 31/08/2024
- Granting institution: Spanish Ministry of Science and Innovation.
  - Duration: 3 years.
  - Amount: €435,500.
  - Principal investigators: Ricardo Cao Abad and Juan Manuel Vilar Fernández.
  - Code: PID2020-113578RB-I00.
  - Role: PhD Student.

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## Peer Review Experience

- *Advances in Data Analysis and Classification* (1 reviewed manuscript).
- *Annals of Applied Statistics* (1 reviewed manuscript).
- *Annals of Operations Research* (2 reviewed manuscripts).
- *Artificial Intelligence Review* (1 reviewed manuscript).
- *Computational Statistics* (3 reviewed manuscripts).
- *Computational Statistics and Data Analysis* (1 reviewed manuscript).
- *Expert Systems with Applications* (1 reviewed manuscript).
- *Fuzzy Sets and Systems* (1 reviewed manuscript).
- *IEEE Access* (1 reviewed manuscript).
- *Information Sciences* (2 reviewed manuscripts).
- *International Journal of Approximate Reasoning* (1 reviewed manuscript).
- *Journal of Applied Statistics* (1 reviewed manuscript).
- *Journal of Computational and Graphical Statistics* (1 reviewed manuscript).
- *Journal of Nonparametric Statistics* (1 reviewed manuscript).
- *Nature* (1 reviewed manuscript).
- *Pattern Recognition* (1 reviewed manuscript).
- *Spatial Statistics* (2 reviewed manuscripts).
- *Statistics and Computing* (1 reviewed manuscript).
- *Stochastic Environmental Research and Risk Assessment* (1 reviewed manuscript).